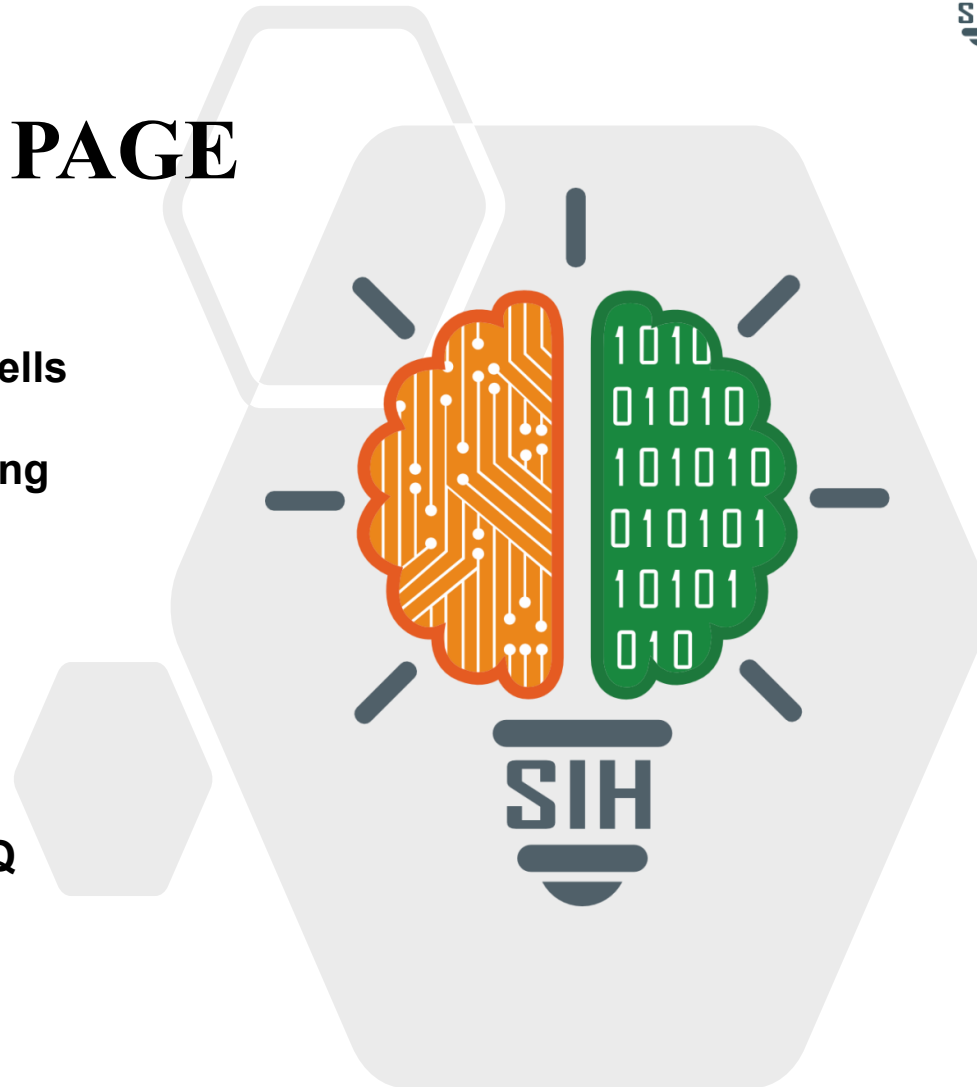


TITLE PAGE

- Problem Statement ID – SIH26121
- Problem Statement Title- eRTMAC-NWIS (Nearby Wells Intelligence System): An AI-Powered Offset Well Knowledge and Decision Support Platform for Drilling Operations
- Theme- Smart Automation
- PS Category- Software
- Team ID-
- Team Name (Registered on portal)
- Organisation- Oil India Limited | eRTMAC, Field HQ Duliajan, Assam

*“What nearby wells learned,
on screen before the bit gets there”*



Your
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NWIS: Offset-Well Memory with Look-Ahead Alerts



Proposed Solution (Describe your Idea/Solution/Prototype)

For Oil India's drilling teams at Duliajan and on the rig: a memory of what went wrong in nearby wells, drawn from their old reports, that warns the rig before the bit reaches the same trouble.

Detailed explanation of the proposed solution



A standalone memory beside the live feed: old well reports become a searchable, cited record of what went wrong.

Reads Well Completion + Daily Drilling Reports (WCR, DDR)

Every extracted incident cites its page + a confidence

Low-confidence rows wait in an engineer review queue

Two views: Duliajan office (full) · rig (alerts first)

Runs beside eRTMAC, Duliajan's real-time drilling centre

How it addresses the problem



Today the answer sits in someone's memory or a report nobody has time to read; now it is one question away.

Scattered PDFs → one structured incident ledger

Nearest well ≠ best offset → ranked by similarity

Wells at different depths → lined up by formation

Alerts after the event → warning before the bit arrives

Hours of manual search → a cited answer in plain words

Innovation and uniqueness of the solution



It looks ahead of the bit, not at it — and shows the report page behind every warning.

Risk per depth, ahead of the bit: we found no prior method

Formation depths corrected for dip and its direction

A missing clue is shown as n/a, never filled with zero

Alarm-fatigue control: dedupe, cooldown, hard cap

Local AI model, offline: reports never leave Oil India

Links (fill in before submission)



Live dashboard:

<https://> _____



Demo video:

<https://> _____



GitHub repository:

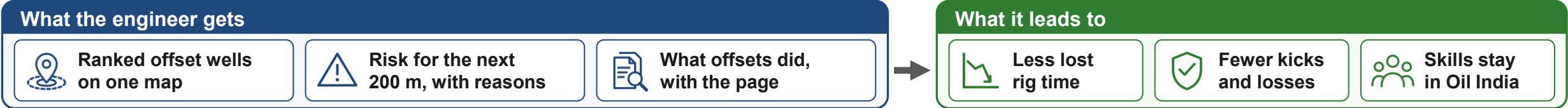
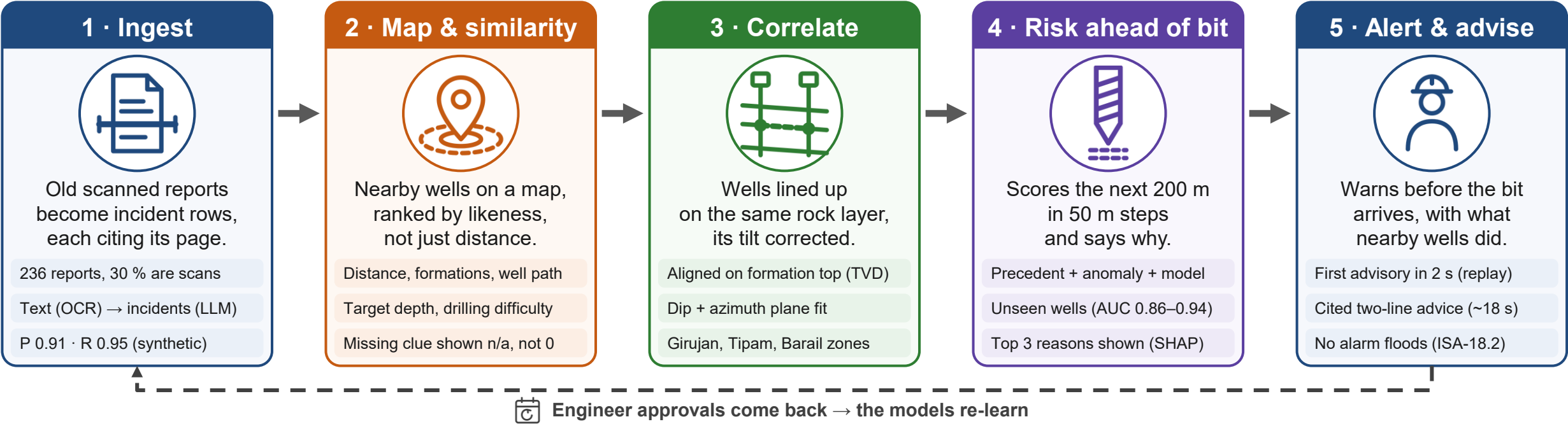
<https://> _____

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TECHNICAL APPROACH



- Technologies to be used: Python · FastAPI · Ollama qwen2.5 · RapidOCR · XGBoost + SHAP · SQLite/PostGIS · React + Leaflet
- Methodology and process for implementation: WCR/DDR PDFs → OCR + local-LLM extraction (cited, confidence-scored) → incident ledger + review queue → similarity map → formation correlation → risk-ahead-of-bit fusion → alerts on the eRTMAC feed



Numbers from a synthetic Upper Assam field (30 wells) · checked on real North Sea reports · offline

Feasibility and viability — a working prototype, its honest gaps, and how each gap closes once Oil India's own reports flow in.

Analysis of the feasibility of the idea



Built and running today —
end to end, offline,
on one laptop.

- All seven asks in the problem statement demoed
- Rock column from Oil India's own paper (SPG 2013)
- Real-report check on Equinor's open Volve data
- Open-source stack · no cloud, no licence cost

Potential challenges and risks



Old paper, shifting formats
and no live feed yet —
the real risks.

- Faded scans and stamps defeat text reading (OCR)
- Report formats drift across decades and fields
- Terse rig shorthand is missed on real reports
- No eRTMAC access yet; new areas lack offsets

Strategies for overcoming these challenges



Every weak spot has a fallback
and an engineer
in the loop.

- Confidence score → doubtful rows go to review
- Per-template reading profiles, not one parser
- Approvals retrain it: ~60 labelled report-days (DrillScribe)
- Feed adapter ready (WITSML); basin priors at cold start

Roadmap



Now: working prototype,
office + rig views



Next: sample Oil India
reports → re-tune



Finale: 36-hour build,
live office + rig demo



Pilot: one Duliajan
rig, advisory mode

Team: Chemical Engineering + Software, MNIT Jaipur · open-source, on-premise, slots into Project DRIVE 2.0 beside eRTMAC

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IMPACT AND BENEFITS



Impact and benefits — what changes for Oil India's drilling engineers, and what we honestly do and do not claim yet.

Potential impact on the target audience



Oil India plans 200+ wells in three years — every one drilled beside old ones.

Offset study: days of reading → minutes

Cited answer in under a minute (measured)

Warning before the bit reaches a flagged zone

Rig view built for a thin satellite link

One version of the truth for office and rig

Benefits of the solution (social, economic, environmental, etc.)



Economic

Every rig hour not lost is money not spent

Less non-productive time (NPT)

Fewer stuck-pipe fishing jobs

No licence fee: open source



Environmental

Fewer losses and kicks, less waste at the well

Less mud lost to the rock

Fewer rig days, less diesel

Fewer sidetracks and re-drills



Social & operational

Experience stays with Oil India when experts retire

Juniors learn from cited cases

English and Hindi interface

Safer crews: fewer kick surprises

What we do not claim

Accuracy is measured on synthetic reports — not on Oil India's archive

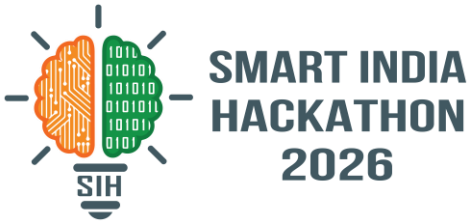
Real Volve reports: few false alarms, but much rig shorthand still missed

No Oil India NPT figure, no eRTMAC access yet; kick risk is an indicator, not a probability

Industry evidence, not our result: Baker Hughes reports offset-well research cut from five days to five hours with automated offset analysis

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RESEARCH AND REFERENCES



Everything we used is public and cited — Oil India's own tenders and papers, open North Sea data, and the drilling-AI work we build on.

Details / Links of the reference and research work

Oil India Ltd — e-RTMAC hiring tender (NIC_CDH6605P21); IDPMS EOI 2023 (eRTMAC feed clause); Project DRIVE 2.0 (oil-india.com/digitalfootprint).	Parmar, H. S. et al. (2025). IPTC-24705-MS: Complex Risk Assessment of Offset Wells Using Cognitive Analysis. IPTC 2025, Kuala Lumpur.
Mandal, K. L. & Dasgupta, R. (2013). Upper Assam Basin and its basinal depositional history. SPG India, 10th Biennial, P292.	Lu, R. (2026). TADI: Tool-Augmented Drilling Intelligence via Agentic LLM Orchestration over Heterogeneous Wellsite Data. arXiv:2605.00060.
Wandrey, C. J. (2004). Sylhet-Kopili/Barail-Tipam Composite Total Petroleum System, Assam Province, India. USGS Bulletin 2208-D.	DrillScribe (2026), open repository: Volve DDR → NPT classifier; cross-operator AUC 0.69 → 0.785 with 60 local labelled report-days.
DGH, MoPNG — National Data Repository (ndr.dghindia.gov.in): India's well completion reports are held largely as unstructured documents.	ANSI/ISA-18.2-2016. Management of Alarm Systems for the Process Industries.
Equinor (2018). Volve field data set: daily drilling reports, WITSML streams and final well reports — Equinor Open Data Licence.	Chen & Guestrin (2016). XGBoost, KDD · Lundberg & Lee (2017). A Unified Approach to Interpreting Model Predictions (SHAP), NeurIPS.
Sodir, Norwegian Offshore Directorate. FactPages: wellbore history, formation tops and coordinates, under the NLOD licence.	Baker Hughes — JewelSuite Offset Well Analyzer ("five days to five hours"); SLB × Dataiku — NLP extraction from daily drilling reports.

Links (fill in before submission)

Live dashboard: https:// _____	Demo video: https:// _____	GitHub repository: https:// _____
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Oil India and Indian sources for the setting · open North Sea data for real reports · full list with links in the repository (research/)